

SOFTWARE ENGINEERING LAB 2

AGILE BACKLOG MANAGEMENT AND SPRINT SIMULATION

Project No.#27: Warehouse Inventory & Pallet Location Tracker

1. Jira backlog with Epics and User Stories.

The top screenshot shows the Jira backlog for the 'Software Engineering Lab' space. It displays two sprints: 'SCRUM Sprint 0' (Sep 2 - 16 Sep) and 'SCRUM Sprint 1' (empty). The backlog is currently empty.

The bottom screenshot shows the Jira backlog for the 'Software Engineering Lab' space. It displays a backlog with 9 work items. The items are as follows:

Item ID	Item Name	Category	Status	Priority	Assignee
SCRUM-9	Scan Pallet Using Barcode/RFID	User Story	To Do	3	Assignee
SCRUM-10	Assign Pallet 3D Location	User Story	To Do	5	Assignee
SCRUM-11	Retrieve Pallet Location Summary	User Story	To Do	3	Assignee
SCRUM-12	Monitor Pallet Inventory	User Story	To Do	5	Assignee
SCRUM-13	Validate Rack Weight Capacity	User Story	To Do	5	Assignee
SCRUM-14	Reject Unsafe Pallet Placement	User Story	To Do	3	Assignee
SCRUM-15	Record Pallet Movement	User Story	To Do	5	Assignee
SCRUM-16	Store Pallet Movement Details	User Story	To Do	5	Assignee
SCRUM-17	View Pallet Movement History	User Story	To Do	3	Assignee

2. Story point assignments

Create a whiteboard to plan your workTry

SCRUM-9

Scan Pallet Using Barcode/RFID

Pallet Identification...

To Do ▾

3

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👤

SCRUM-10

Assign Pallet 3D Location

Pallet Identification...

To Do ▾

5

⬇

👤

SCRUM-11

Retrieve Pallet Location Summary

Pallet Identification...

To Do ▾

3

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SCRUM-12

Monitor Pallet Inventory

Pallet Identification...

To Do ▾

5

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SCRUM-13

Validate Rack Weight Capacity

Safe Pallet Storage

To Do ▾

5

⬇

👤

SCRUM-14

Reject Unsafe Pallet Placement

Safe Pallet Storage

To Do ▾

3

⬇

👤

SCRUM-15

Record Pallet Movement

Stock Movement T...

To Do ▾

5

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👤

SCRUM-16

Store Pallet Movement Details

Stock Movement T...

To Do ▾

5

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👤

SCRUM-17

View Pallet Movement History

Stock Movement T...

To Do ▾

3

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👤

3.Sprint board (Active Sprint view)

Sprint 1

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Jira

Search

+ Create

Ask Rovo

Software Engineering Lab

Summary Backlog Board Docs Development

Search board Filter Group

Complete sprint

To Do 0 In Progress 0 Done 4

+ Create

Scan Pallet Using Barcode/RFID

Parent Pallet Identification & Location Ma...

SCRUM-9 3 ✓ ⬇ 👤

Validate Rack Weight Capacity

Parent Safe Pallet Storage

SCRUM-13 5 ✓ ⬇ 👤

Assign Pallet 3D Location

Parent Pallet Identification & Location Ma...

SCRUM-10 5 ✓ ⬇ 👤

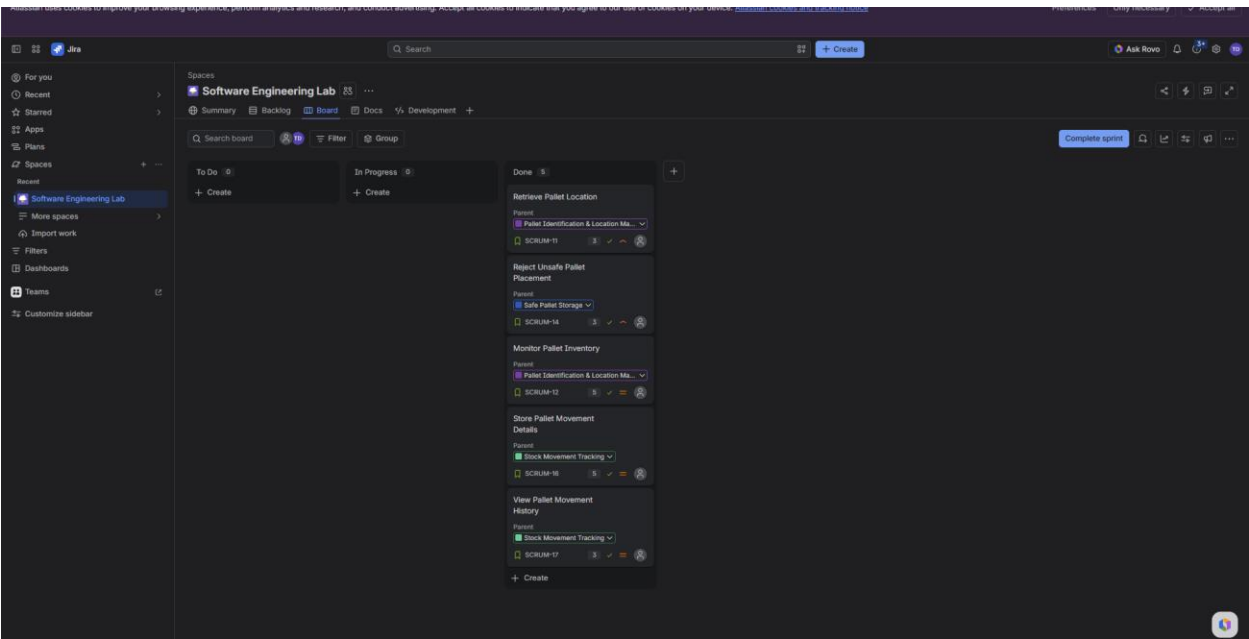
Record Pallet Movement

Parent Stock Movement Tracking

SCRUM-15 5 ✓ ⬇ 👤

+ Create

Sprint 2

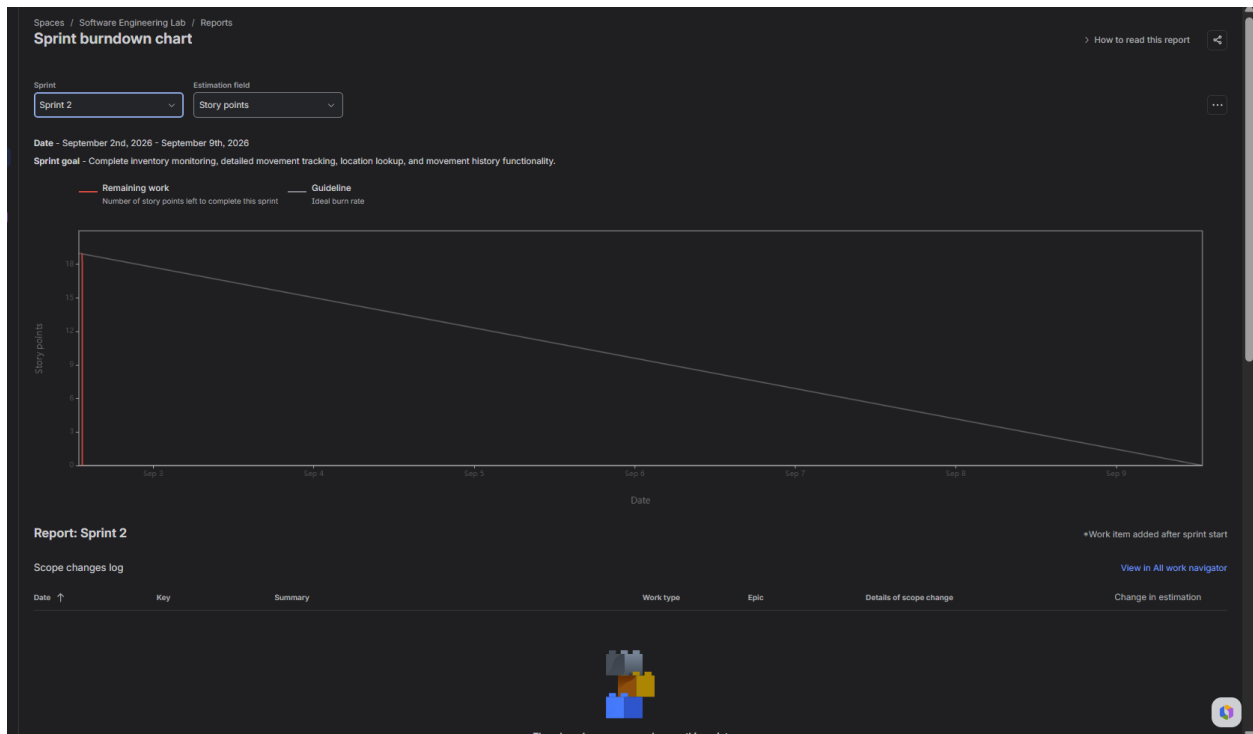


4.Burndown Chart

Sprint 1



Sprint 2



Reflection Questions

Q1. Were your estimations accurate?

The story point estimations generally reflected the relative effort required for the user stories. Smaller and more straightforward stories were assigned 3 story points, while stories involving more complex functionality such as location assignment, rack weight validation, and movement tracking were assigned 5 story points.

Q2. How did you prioritize your backlog?

The backlog was prioritized based on the importance of the warehouse's core functionality. High-priority stories included pallet identification, rack weight validation, pallet location assignment, and stock movement tracking because these functions are essential to the operation of the system. Medium-priority stories such as inventory monitoring and movement history were planned after the core functionality.

Q3. How did you plan and execute the two sprints?

Sprint 1 focused on the core warehouse functionality, including pallet identification, rack weight validation, three-dimensional location assignment, and movement

recording. Sprint 2 focused on location lookup, inventory monitoring, unsafe placement rejection, detailed movement tracking, and movement history. The work items were simulated through the To Do, In Progress, and Done workflow.

Q4. What did the Burndown Chart tell you?

The Burndown Chart showed the amount of remaining work during each sprint based on story points. It provided a visual representation of sprint progress and showed that the planned work was completed within the simulated sprint period. The chart also helped in understanding the relationship between remaining work and the ideal burn rate.